Physically Interpretable World Models for Partially Observable Driving: Encoding and Predicting the Road Environment under Weak Supervision

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Abstract

Learning predictive models from high-dimensional observations is fundamental for safe autonomous cyber-physical systems, yet the latent states of standard world models lack physical interpretability. Our conference work introduced the Physically Interpretable World Model (PIWM), which aligns latent representations with physical quantities and constrains their evolution through partially known dynamics under distribution-based weak supervision. That formulation assumes the vehicle state is a complete Markov description of the task, which holds for fully observable benchmarks such as CartPole and Lunar Lander but fails for driving: two vehicles with identical position, heading and speed follow divergent futures when the road ahead differs, and a forward-facing camera reveals that road only partially. This article extends PIWM to partially observable driving along three axes. First, we move to two partially observable driving case studies: a simulated top-down *CarRacing* track and a *real, physical DonkeyCar*. Second, we introduce a physically interpretable *encoding* of the road environment, factorizing the interpretable latent into a vehicle-relative component and a road-context component—lateral deviation, vehicle–road heading error and curvature—inferred from onboard images. Third, we introduce a physically interpretable *prediction* of that environment: a split-dynamics model that advances the vehicle through structured kinematics while forecasting how the road context evolves. On the physical DonkeyCar, where road context is supervised only during training and must be recovered from the front camera at test time, PIWM attains 0.463 m position error over a 100-step rollout against 0.703 m for the strongest baseline under five-fold cross-validation, ahead in every fold, with less than half the fold-to-fold variation of any baseline—and with every state variable physically meaningful.

Keywords: physically interpretable world models, partial observability, autonomous driving, road-context encoding, weak supervision, trajectory prediction

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1. Introduction

Accurate and robust trajectory prediction from high-dimensional sensor data is a foundational challenge for autonomous cyber-physical systems (CPS). Modern *world models* [1] address this challenge by encoding observations into compact latent representations and predicting their temporal evolution, a paradigm that has proven effective for long-horizon prediction, planning, and control [2, 3, 4]. Despite strong empirical performance, the latent states of these models typically lack a clear connection to the underlying physical state of the system. In safety-critical settings such as autonomous driving and robotics, this opacity limits trustworthiness: without interpretable states, it is difficult to detect unsafe future configurations, diagnose prediction failures, or apply formal safety-monitoring techniques [5, 6, 7].

Our prior conference work introduced the Physically Interpretable World Model (PIWM) [8], a framework that aligns latent representations with physical quantities and constrains their temporal evolution through a partially known dynamics model. PIWM defines physical interpretability through two complementary properties: the latent state must correspond to meaningful physical variables (state semantics), and its temporal evolution must follow physically consistent dynamics (temporal correctness). A key feature of PIWM is *distribution-based weak supervision*: rather than requiring precise ground-truth state labels, the method accepts state estimates in the form of sample distributions, as naturally produced by real-world sensing pipelines such as GPS or radar. Experiments on CartPole, Lunar Lander, and DonkeyCar demonstrated that PIWM recovers accurate physical parameters and achieves long-horizon prediction accuracy well beyond data-driven baselines. A systematic comparison of intrinsic versus extrinsic architectures and continuous versus discrete latent spaces showed that an extrinsic architecture with a quantized (VQ-based) latent yields the most robust and physically grounded representations.

The challenge: driving is partially observable

Applying PIWM to realistic driving exposes a fundamental limitation of the original formulation, and motivates the extensions in this article. On CartPole and Lunar Lander, the system state completely specifies what will happen next: the dynamics are fully observable and Markov, so a world model that tracks the vehicle (or pole, or lander) state captures everything relevant to the future. Driving is different. Consider two vehicles with identical position, heading, and speed. If one is approaching a tight curve while the other is on a straight segment, their future trajectories diverge even under the same control inputs. The vehicle's physical state $x_t = (x, y, \psi, v)$ does not capture this difference. The missing information is the *road context*: the shape of the lane ahead (its curvature) and the vehicle's lateral and angular relationship to it, none of which is part of any standard vehicle-state representation. As a result, the vehicle state alone is *non-Markov* with respect to future motion: two identical states lead to different futures depending on the unobserved road geometry. Applying the original PIWM formulation to such tasks implicitly treats the system as fully observable when it is not, causing the dynamics model to learn an underdetermined, ill-posed mapping.

This challenge has two distinct facets, and overcoming each requires a separate mechanism. *(i) The road environment must be encoded.* The relevant road context is not given as a clean state vector; it is latent in the pixels and, for a forward-facing camera, only partially visible at any instant. A physically interpretable driving world model must therefore *infer* the geometry of the surrounding road from raw images, and represent it in physically meaningful terms rather than as opaque features. *(ii) The road environment must be predicted.* Long-horizon rollout requires anticipating how the road context itself evolves as the vehicle advances: the lane that is far ahead now becomes the lane the vehicle is on a few steps later. A world model that predicts only the vehicle state, treating the road as static or as noise, accumulates error rapidly because it cannot anticipate the geometry it is about to enter. Predicting the road, like encoding it, should be physically interpretable so that the forecasted environment can be inspected, trusted, and monitored.

These two facets, encoding and predicting the external environment, are what distinguishes a driving world model from the vehicle-centric formulation of the conference paper, which modeled only the state of the controlled body. They are also why partial observability cannot be papered over by simply adding capacity to a black-box predictor: the missing variable is structured, physical, and causally upstream of the vehicle's future, so it deserves an explicit, physically grounded treatment.

Our approach

We resolve partial observability by reformulating the interpretable state as a structured composition of two complementary components. The *vehicle-relative state* z^v captures the ego vehicle's displacement and heading change expressed in a local reference frame anchored at the beginning of each prediction window; this representation removes dependence on global coordinates and improves transferability across trajectories. The *road-context state* z^r captures task-relevant environmental structure: the cross-track error (lateral deviation from the lane center), the vehicle–road heading error, and the local road curvature. Together, (z^v, z^r) form a Markov description of the driving task, restoring the validity of the predictive world-model framework. The vehicle-relative component is governed by a structured kinematic model whose unknown actuation parameters are recovered during training; the road-context component is governed by a geometry-driven transition that advects the road past the vehicle as it advances, rather than treating the road as a free latent that can drift. This split-dynamics formulation provides appropriate inductive biases for each component and decouples the faster-varying vehicle motion from the slower-varying road context.

We validate the extended formulation in two new partially observable driving environments. In **CarRacing**, a top-down simulated racing environment, the road geometry is present in the image but absent from the vehicle state; the model must learn to infer and exploit road-context information from raw pixels. On the physical **DonkeyCar** platform, a real RC-scale car observes its environment through a front-facing camera and drives on a laboratory track. During training, an external localization system provides ground-truth road context as a privileged signal; at test time this signal is withheld, and the model must infer and roll out road context solely from the onboard front-facing image stream. This setup tests whether the factorized representation learned under privileged supervision transfers to the more challenging unaided real-world deployment setting. On the physical platform, over a 100-step ($\approx$4.5 s) rollout and under five-fold cross-validation, PIWM with factorized state and split dynamics reaches 0.463 m position error against 0.703 m for the strongest data-driven or physics-informed baseline, ahead in every fold, and the gap widens with the horizon: the baselines diverge while PIWM stays bounded.

Contributions

This journal article extends the conference version [8] along three axes, which constitute its three contributions.

1. **Two partially observable driving case studies.** We move PIWM beyond the fully observable benchmarks of the conference paper to two new driving environments that are genuinely partially observable: a simulated top-down *CarRacing* track and a *real, physical DonkeyCar* platform driven on a laboratory track under onboard-camera observation. We formally characterize partial observability in driving, showing that the vehicle state is non-Markov with respect to future trajectory, and use these two case studies to demonstrate the practical consequences and the benefit of the proposed treatment.
2. **Physically interpretable encoding of the road environment.** Unlike the conference formulation, which modeled only the controlled body's state, we model the *external environment*. We introduce a factorization of the interpretable latent into a vehicle-relative component z^v and a road-context component z^r, where z^r comprises physically meaningful descriptors of the road (cross-track error, vehicle–road heading error, and curvature). We show how to infer z^r from images using temporal context, including the forward-camera case where the road ahead is only partially visible, and how privileged supervision available only at training time can be used to learn this encoding.
3. **Physically interpretable prediction of the road environment.** We introduce a split-dynamics model that not only rolls out the vehicle with a structured kinematic model, but also *forecasts the road context* with a geometry-driven transition in which the road is advected past the vehicle by its own predicted motion. The predicted road context is itself physically interpretable, so the forecasted environment, not just the forecasted vehicle, can be inspected and monitored. We show that this geometric treatment, rather than propagating the road as a free latent, is what keeps long-horizon rollout bounded, and that the road context can be perceived from the onboard camera at a cost of roughly 2 cm in 100-step position error relative to a privileged map lookup.

The remainder of this article is organized as follows. Section 2 surveys related work. Section 3 reviews background on autonomous CPS, world models, and physical interpretability. Section 4 formalizes the partial observability problem in driving and introduces the factorized state. Section 5 presents the PIWM architecture, the road-context encoder (Section 5.2), and the road-context predictor (Section 5.3). Section 6 describes the experimental setup, and Section 7 reports results on the benchmarks and on the real DonkeyCar. Section 8 discusses implications and limitations, and Section 9 concludes.

2. Related Work

We organize the discussion into five threads relevant to physically interpretable world modeling under partial observability: trajectory prediction and safe forecasting (Section 2.1), representation learning for dynamical systems (Section 2.2), world models (Section 2.3), physics-informed and interpretable learning (Section 2.4), and partial observability, privileged learning, and road structure (Section 2.5). We close each thread by positioning PIWM against it.

2.1. Trajectory Prediction and Safe Forecasting

Predicting future motion is central to safe planning and control [9]. Classical predictors propagate a low-dimensional state with hand-specified models: constant-velocity and Kalman-filter formulations for vehicle motion [10], and Hamilton–Jacobi reachability for sets of reachable states with formal guarantees [11, 12, 13]. These methods are transparent and certifiable, but they require an explicit dynamical model and a low-dimensional state, and they scale poorly to high-dimensional camera observations.

Learning-based predictors instead forecast directly from data. Social-pooling and generative sequence models capture multi-agent interaction [14, 15, 16], while graph- and attention-based models such as Trajectron++ achieve strong forecasting by conditioning on scene graphs and high-definition maps [17]. Hybrid schemes embed classical filters inside learned models, as in learned Kalman filtering [18]. These models, however, depend on handcrafted scene representations or HD-map annotations that are unavailable in many settings [19, 20], and their internal states carry no physical meaning.

A parallel line equips predictors with reliability guarantees without altering the predictor itself: conformal prediction and calibrated forecasting bound error or coverage [21, 22, 23, 24]; statistical model checking and predictive monitoring estimate the probability of property violations [25, 26, 27, 28]; and runtime and misbehavior monitors flag unsafe behavior online [29, 30, 31, 32]. These techniques are complementary to ours: they wrap a predictor with guarantees but do not make its latent state physically meaningful. Compared with this body of work, PIWM learns interpretable representations directly from raw images under weak supervision, without handcrafted scene representations or test-time state labels, so that the forecast itself, both the vehicle and the road around it, can be inspected.

2.2. Representation Learning for Dynamical Systems

Compact latent representations underpin learning from high-dimensional observations, from classical autoencoders [33] to variational autoencoders (VAEs) [34], but standard objectives impose no physically meaningful structure. A large literature seeks more structured latents by encouraging disentanglement [35, 36, 37], causal structure [38, 39], or object-centric and slot decompositions [40]. Discrete VQ-VAEs regularize the latent through a learned codebook [41, 42]; PIWM builds on this quantization but, unlike it, ties codebook components to physical values. These methods improve organization or modularity, yet none guarantees correspondence to measurable physical quantities.

Closer to physical grounding, several models steer latents toward physics. GOKU-net constrains the latent dynamics to a known ODE and infers that ODE's parameters [43], but the functional form must be fixed a priori and the latents are tied to it rather than to quantities that can be measured from the observation; SPARTAN produces semantically structured outputs but incorporates neither known dynamics nor action-conditioned prediction [44]; Deep Variational Bayes Filters add latent state-space dynamics to VAEs [45] but, absent physical supervision, do not recover interpretable variables; and continuous-time models such as Neural ODEs learn smooth but uninterpretable latent dynamics [46]. Representation learning specialized for pixel-based control improves downstream performance [47] while again leaving the latent opaque. Compared with these approaches, PIWM explicitly aligns latent dimensions with named physical quantities of both the vehicle and the surrounding road, and constrains their evolution with known physics under only weak, distribution-based supervision.

2.3. World Models

World models learn latent dynamics for imagination-based planning and control. The Dreamer and PlaNet line learns recurrent latent state-space models from pixels [48, 49, 3]; transformer- and masked-prediction world models improve sample efficiency and long-horizon fidelity [2, 4, 50]; and a range of variants explore alternative latent factorizations [1, 51, 52]. These models excel empirically but learn latent spaces with no clear physical meaning, which limits inspection and safety monitoring.

For driving, recent world models predict rich structured futures such as 3D occupancy grids or rendered scenes [53, 54, 55, 56], improving forecasting realism but still without aligning the internal state to physical variables, and at substantial annotation and compute cost. Neuro-symbolic world models improve generalization but require predefined symbolic inputs unavailable from raw sensors [57, 58]. The closest prior work, Vid2Param, recovers physical parameters from video [59] but requires full supervision and provides no explicit world model for long-horizon rollout. Compared with these approaches, PIWM recovers physical state under weak supervision and rolls it forward through interpretable dynamics, and, new in this article, it does so for the external road environment, not only the controlled body.

2.4. Physics-Informed and Interpretable Learning

A complementary thread injects physics directly into learning. Sparse identification of nonlinear dynamics and differentiable-physics or end-to-end system identification recover governing equations or parameters [60, 61, 62], while physics-informed neural networks embed differential constraints in the loss [63, 64]. Within world modeling, physical priors have been added as bounds on states and actions [65, 66], physics-aware losses [67], kinematics-inspired layers [68], and Taylor-monomial-structured latent dynamics (Phy-Taylor) [69].

Most of these methods assume access to the underlying low-dimensional state and are not designed for raw visual inputs, or they enforce physical plausibility without yielding a state that names specific quantities. PIWM differs in two ways aligned with this article's contributions: it extracts the physical state from images under weak supervision, so that physics can be applied without ground-truth labels, and it applies a structured, learnable physical model, a bicycle model for the vehicle and a geometry-driven transition for the road, to roll the interpretable state forward. Compared with physics-informed predictors that model only the controlled body, PIWM applies interpretable physical structure to the external environment as well.

2.5. Partial Observability, Privileged Learning, and Road Structure

Partial observability is a long-standing challenge in robotics and reinforcement learning, classically formalized through partially observable Markov decision processes and belief-state planning [70, 71]. Model-free agents commonly cope with it by adding memory through recurrent or temporal architectures: recurrent policy gradients [72], gated recurrent networks [73], convolutional LSTMs for spatiotemporal inputs [74], and temporal convolutional networks [75]. These approaches recover an opaque belief or memory state; they do not produce a physically interpretable estimate of the missing environmental variables, which is exactly what our road-context state provides.

When information needed for learning is available only at training time, privileged-information and teacher–student learning supervise a model that must operate without it at deployment [76, 77]. This is the mechanism we use on the real DonkeyCar: ground-truth road context from training-time track localization teaches the encoder what to extract, while the deployed car observes only its front camera and must reconstruct that context from images.

In driving specifically, a substantial literature encodes lane and road structure in learned representations [17, 19], but typically requires explicit maps or HD annotations; and physics-based predictors apply kinematic or dynamic bicycle models to low-dimensional state vectors [78] rather than to learned image representations. Learning-based control has also been demonstrated on RC-scale physical cars, including imitation and reinforcement learning on platforms akin to the DonkeyCar [79, 80], but without an interpretable predictive world model. Compared with all of the above, our work is, to our knowledge, the first to integrate a factorized vehicle–road representation with a physics-informed world model trained from images under weak supervision, and to both *encode* and *predict* the road environment in physically interpretable terms.

3. Preliminaries

3.1. Autonomous CPS and World Models

Definition 1 (Autonomous CPS)**.** *An* autonomous CPS $s = (X, I, Y, A, \phi_\theta, g, h)$ *models the evolution of an observation-based control system, where X is the physical state space, $I \subset X$ the set of initial states, Y the observation space, A the action space, $\phi_\theta : X \times A \times \Theta \to X$ the dynamics with unknown parameters $\theta \in \Theta$, $g : X \to Y$ the observation function, and $h : Y \to A$ the controller.*

The true state evolves as $x_{t+1} = \phi(x_t, a_t, \theta^*)$, where θ^* are the unknown true parameters. The controller selects actions from observations: $a_t = h(y_t) = h(g(x_t))$. A world model serves as a learned surrogate for forecasting future observations.

Definition 2 (World Model)**.** *A world model $\mathcal{W} = (\mathcal{E}, f, \mathcal{D})$ predicts the future evolution of observations in an autonomous CPS s, where $\mathcal{E} : Y \to Z$ encodes observations into latent representations, $f : Z \times A \to Z$ predicts future latents, and $\mathcal{D} : Z \to Y$ reconstructs predicted observations.*

Conventional world models are evaluated by predictive accuracy in image space (e.g., MSE or SSIM). However, low image error does not imply that the underlying predicted state is physically meaningful: similar-looking frames can correspond to very different physical configurations. For safety monitoring and planning, what matters is not only visual fidelity but whether the latent state can be interpreted as a physical quantity and evolved according to physically plausible dynamics.

3.2. Physical Interpretability

Following the conference paper [8], we define *physical interpretability* via two complementary properties:

1. the latent representation z^* corresponds to meaningful physical variables (state semantics), and
2. its temporal evolution follows physically consistent dynamics (temporal correctness).

Property 1 requires alignment between the latent and the physical state, which PIWM enforces through weak supervision. Property 2 requires that multi-step rollouts of the latent state obey the known equations of motion, which PIWM enforces through the structured dynamics model.

Definition 3 (Interpretable Representation Learning Problem)**.** *Given a dataset $\mathcal{S} = \{(y_{1:M}, a_{1:M}, \Xi_{1:M})\}_{1:N}$, where $\Xi_t = \{\xi_t^{(l)}\}_{l=1}^{L}$ are samples drawn from an unknown supervisory distribution $p(x_t)$, and a controller h, learn a latent encoder $\mathcal{E}$ such that the mean squared error between the physical state x_t and the encoded latent $z_t^* = \mathcal{E}(y_t)$ is minimized:*

$$\min_{\mathcal{E}} \ \mathbb{E}\Big[\|x_t - \mathcal{E}(y_t)\|_2^2\Big]. \tag{1}$$

The supervisory samples Ξ_t are drawn from an unknown distribution $p(x_t)$ with no assumed parametric form; we only require that the distribution is *mean-proximal*, meaning its mean lies within a relative interval of width δ around the true value (see Section 6.2 for the precise noise model). This is a strictly weaker assumption than standard supervised learning with precise labels, and represents the uncertainty naturally arising from real-world sensing.

4. Partial Observability in Driving

This section formalizes the challenge introduced in Section 1: in driving, the vehicle state is not a sufficient statistic for the future. The formalization motivates the factorized state that the rest of the article encodes (Section 5.2) and predicts (Section 5.3).

4.1. Why Vehicle State Is Insufficient for Driving

The PIWM conference paper formulated physical interpretability under the implicit assumption that the vehicle's physical state x_t provides a complete, Markov description of the system: knowing x_t and action a_t determines x_{t+1}. This assumption holds for CartPole, Lunar Lander, and other low-dimensional benchmark tasks, where the system state encodes all task-relevant information. It fails for driving tasks.

In a driving task, the vehicle state $x_t = (x, y, \psi, v)$ (position, heading, speed) does not encode the geometry of the road ahead. The actual lane-following dynamics of the vehicle depend on this geometry:

$$x_{t+1} = \phi(x_t,\ r_t,\ a_t,\ \theta^*), \tag{2}$$

where r_t denotes the *road-context state*, which includes quantities such as the lateral deviation from the lane center, the vehicle's heading relative to the road tangent, and the local curvature. The road-context component r_t is not directly observed by the onboard camera and is not part of any standard vehicle-state representation.

Definition 4 (Partial Observability in Driving)**.** *A driving task is* partially observable *with respect to world modeling if the vehicle state x_t alone is not sufficient to determine future motion; equivalently, if there exist two system configurations $(x_t, r_t^{(1)})$ and $(x_t, r_t^{(2)})$ with $r_t^{(1)} \neq r_t^{(2)}$ such that, under the same controller h, the resulting trajectories satisfy $x_{t+1:T}^{(1)} \neq x_{t+1:T}^{(2)}$.*

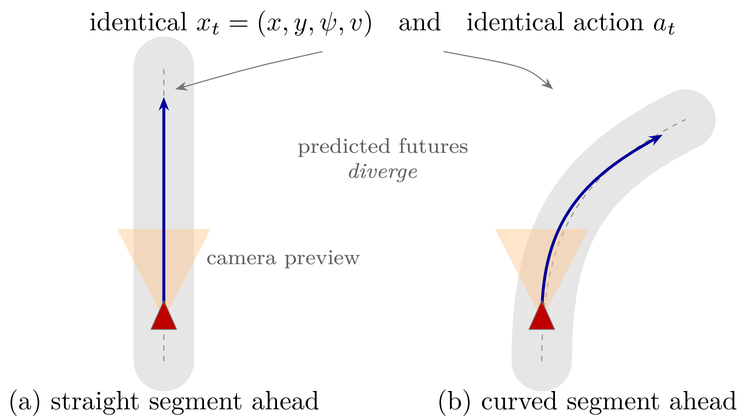

Figure 1: Partial observability in driving. Two vehicles with identical kinematic states (x, y, ψ, v), issued identical actions, follow divergent futures because the road geometry ahead differs. The vehicle state alone is therefore non-Markov: the road context r_t (lane shape and the vehicle's relation to it) is missing from it and, for a forward-facing camera, the road is visible only over a short preview (shaded wedge), so r_t must be both encoded from images and predicted forward.

Figure 1 illustrates this definition. Two vehicles at identical positions on the track, with the same speed and heading, enter different road segments (one straight, one curved). Despite identical vehicle states and identical control actions, the predicted future trajectories diverge, because the curvature of the upcoming road introduces a lateral displacement that the vehicle-state-only dynamics cannot anticipate. Standard PIWM, which learns a single interpretable state aligned with x_t, cannot represent this difference. Two consequences follow, corresponding to the two facets of the challenge from Section 1: the model must *encode* r_t from observations (Section 5.2), and it must *predict* how r_t evolves over the horizon (Section 5.3).

4.2. Factorized Interpretable State

To resolve the partial observability problem, we augment the interpretable state with road-context information and express the vehicle component in a translation- and rotation-invariant reference frame.

Vehicle state. Global coordinates (x, y) introduce unnecessary dependence on the starting position of each trajectory, complicating generalization across scenes. In the road-aligned frame the vehicle's own motion is described by

$$z_t^v = \begin{bmatrix} v_t \\ \omega_t \end{bmatrix}, \tag{3}$$

its longitudinal speed and yaw rate, both independent of where the trajectory started.

Road-context state. The road-context component z_t^r captures the vehicle's configuration relative to the road geometry:

$$z_t^r = \begin{bmatrix} s_t \\ e_{\mathrm{CTE}} \\ e_\psi \end{bmatrix}, \tag{4}$$

where s_t is the arc length travelled along the reference centerline, e_{CTE} is the cross-track error (signed lateral distance from the lane center) and e_ψ is the heading error (difference between the vehicle's heading and the road tangent at its current position). Locating the vehicle by s_t along the road rather than in the world is what allows the road description below to be indexed by it.

Factorized state. The complete *factorized interpretable state* is

$$z_t^* = [z_t^v,\ z_t^r] = (v_t,\ \omega_t,\ s_t,\ e_{\mathrm{CTE},t},\ e_{\psi,t}) \in \mathbb{R}^5. \tag{5}$$

The local curvature κ is deliberately *absent* from (5): it is not propagated. The model instead carries a curvature profile $\hat{\kappa}(\cdot)$ over arc length, perceived once from the image window at the start of the rollout, and reads the curvature at the vehicle from it,

$$\kappa_t = \hat{\kappa}(s_t - s_{t_0})\,. \tag{6}$$

This distinction is what the rest of the article rests on. A model carrying curvature as a state variable must predict how it evolves, and that prediction compounds over a long rollout; re-indexing a profile observed once cannot compound, because only the vehicle's position along the road advances.

The Markov property therefore holds for the pair $(z_t^*,\ \hat{\kappa}(\cdot))$ with $\hat{\kappa}$ fixed across the rollout: given the current factorized state, the perceived profile and the action, the next factorized state is determined up to the unknown parameters θ^*. It does not hold for z_t^* alone, since (6) requires the profile.

4.3. Extended Learning Problem

The learning problem for partially observable driving extends the original PIWM formulation as follows.

Problem 1 (Interpretable State Learning under Partial Observability)**.** *Given a dataset* $\mathcal{S} = \{(y_{1:M}, a_{1:M}, \Xi^v_{1:M}, \Xi^r_{1:M})\}_{1:N}$ *where* Ξ^v_t *and* Ξ^r_t *are supervisory samples for the vehicle and road-context components respectively, and a controller* h*, learn a factorized physical encoder* $\mathcal{E}_p$ *whose two outputs* $(z^v_t, z^r_t) = \mathcal{E}_p(y_{1:t})$ *approximate the vehicle state and the road-context state, and which supports accurate future prediction under the task dynamics.*

Two aspects distinguish this from the conference formulation, and they map directly onto the two extensions developed next. First, the encoder is conditioned on a *history* $y_{1:t}$ rather than a single frame: road context cannot be inferred from a single image (particularly in the front-camera setting) and requires temporal context (Section 5.2). Second, the supervision is split: Ξ^v_t can often be computed directly from the vehicle state (available in simulation or via external localization), while Ξ^r_t requires knowledge of the road geometry. In the real DonkeyCar experiments, road-context supervision is available only at training time and withheld at test time, so the encoder must learn to recover road context from images alone (Section 5.2), and the predictor must roll it forward (Section 5.3).

5. PIWM Architecture

The PIWM architecture retains the core structure from the conference version [8], comprising a physically interpretable autoencoder and a learnable dynamics model, and extends it with the factorized state along two axes: a road-context *encoder* (Section 5.2) and a road-context *predictor* (Section 5.3). A high-level overview is shown in Figure 2.

5.1. Learning Interpretable Representations

The autoencoder maps high-dimensional observations to a physically interpretable latent state aligned with the true physical state. We follow the conference paper in considering two architectural approaches.

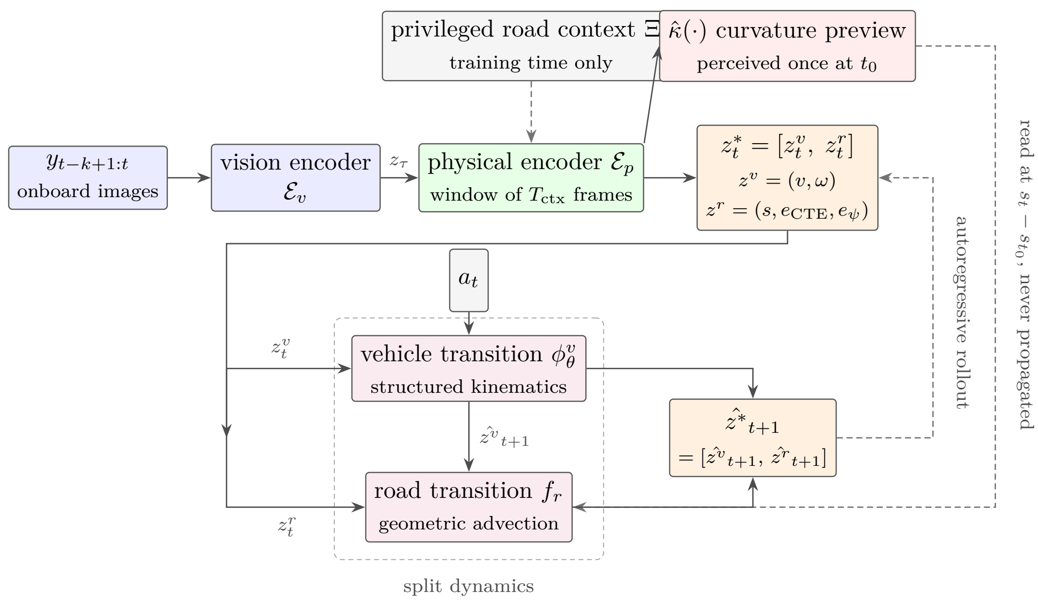

Figure 2: Overview of the PIWM architecture for partially observable driving. A vision encoder maps each observation to a visual latent; a physical encoder aggregates a window of those latents into the factorized interpretable state $z^* = [z^v, z^r]$ and, separately, into a curvature preview $\hat{\kappa}(\cdot)$ over the road ahead. Road-context supervision is privileged and available at training time only. A split-dynamics model then rolls the state forward: z^v by a structured kinematic model, z^r by a geometric one. Two couplings carry the article's argument. The road advances only through $\hat{z^v}_{t+1}$, advected by the vehicle's own predicted motion, which is why the action enters the vehicle branch alone. And the preview is *read* at the arc length covered rather than propagated (dashed, right), so no error in the road can compound over the rollout.

Intrinsic encoding. The encoder $\mathcal{E} : Y \to Z^*$ maps an observation directly to the interpretable latent, partitioned as $z^* = [z_p^*, z_{\mathrm{app}}^*]$, where z_p^* is the physically interpretable component and z_{app}^* captures the residual appearance information needed for reconstruction. The conference version writes the latter z_v^*; we rename it here because this article uses the superscript v for the *vehicle* component of the factorized state. For continuous latent spaces, the objective follows a β-VAE formulation with interpretability alignment applied to z_p^*:

$$\mathcal{L}_{\text{intrinsic-cont}} = \mathcal{L}_{\text{recon}}(y, \hat{y}) + \lambda_{\text{interp}}\, \mathcal{L}_{\text{interp}}(z_p^*, \Xi) + \beta\, D_{\mathrm{KL}}\Big(q(z_{\mathrm{app}}^*|y) \,\|\, \mathcal{N}(0, I)\Big)\,. \tag{7}$$

For discrete latent spaces, the codebook is partitioned so that each entry $\mathbf{e}_k = [\mathbf{e}_k^p, \mathbf{e}_k^v]$ has a fixed physical component and a learnable visual component. The interpretability loss is applied to the averaged physical part of the selected codebook entries.

Extrinsic encoding. The extrinsic approach decouples visual perception from physical interpretation via a two-stage process. In the first stage, a vision autoencoder $(\mathcal{E}_v, \mathcal{D}_v)$ is trained on image reconstruction only, producing an intermediate latent $z = \mathcal{E}_v(y)$. For continuous variants this is a β-VAE; for discrete variants a VQ-VAE with a 512-entry codebook. In the second stage, the vision encoder is frozen and a physical encoder $\mathcal{E}_p$, paired with a physical decoder $\mathcal{D}_p$ that maps the interpretable state back to the intermediate latent, is trained to produce $z^* = \mathcal{E}_p(z)$, optimizing:

$$\mathcal{L}_{\text{physical}} = \lambda_{\text{interp}}\, \mathcal{L}_{\text{interp}}(z^*, \Xi) + \lambda_{\text{latent}}\, \mathcal{L}_{\text{recon}}(z, \mathcal{D}_p(z^*))\,. \tag{8}$$

The physical encoder $\mathcal{E}_p$ maps the temporal latent sequence to the factorized state $z^* = [z^v, z^r]$ (Section 5.2).

Which encoder for driving. The conference paper's systematic comparison found that the extrinsic approach with a discrete (VQ) latent gave the strongest physical grounding and prediction accuracy, and it would be natural to carry that configuration over here. We do not, and the reason is worth stating rather than glossing: that comparison was run on *fully observable* benchmarks, where the quantity to be encoded is the controlled body's own state and a single frame nearly determines it. The road-context encoder faces a different problem—reading the geometry of the scene ahead out of a short, partially informative image window—so the conference ranking is an extrapolation, not a result. We therefore tested it directly, implementing the claimed configuration (per-frame convolutional encoder, a 512-entry VQ bottleneck, and a Transformer with self-attention over the temporal window) at matched capacity and training it to convergence on the same data as the encoder we ship. It is worse at the perception task (RMSE_κ 0.219 against $0.154\,\text{m}^{-1}$) and indistinguishable downstream: paired within folds, its 100-step position error differs from the shipped encoder's by less than a millimeter (Table 2). The driving experiments therefore use a convolutional encoder over the stacked window with a linear curvature head, trained from scratch on the road-context objective; the VQ–Transformer variant is reported as a baseline. Section 7.3 returns to why the two differ so much less downstream than they do at perception.

Interpretability loss. For all variants, alignment between the encoded state and the physical variables is enforced through the mean squared error to the empirical mean of the supervisory samples:

$$\mathcal{L}_{\text{interp}}(z^*, \Xi) = \|z^* - \hat{\mu}_\Xi\|_2^2, \quad \hat{\mu}_\Xi = \frac{1}{L}\sum_{l=1}^{L} \xi^{(l)}. \tag{9}$$

When applied to the factorized state, this loss is evaluated separately on the vehicle-relative and road-context components with supervision from Ξ^v and Ξ^r:

$$\mathcal{L}_{\text{interp}}^{\text{drive}} = \lambda_v\, \mathcal{L}_{\text{interp}}(z^v, \Xi^v) + \lambda_r\, \mathcal{L}_{\text{interp}}(z^r, \Xi^r). \tag{10}$$

The separate weighting of the road-context term λ_r is what forces the encoder to commit capacity to the external environment rather than only to the vehicle.

5.2. Encoding the Road Environment

Challenge and motivation. The conference model encoded only the controlled body's state, which for the benchmarks was directly inferable from a single frame. Road context is different in two ways. First, it is a property of the *environment*, not the agent, so it must be read off the scene rather than the vehicle's appearance. Second, for a forward-facing camera it is only *partially visible*: at any instant the camera sees a limited stretch of road, and quantities such as the curvature of the lane the vehicle is about to enter are not fully determined by the current frame. A single-frame encoder is therefore insufficient; the road context must be aggregated over time.

Temporal road-context encoder. We condition the physical encoder on a history of visual latents:

$$(z_t^v,\ z_t^r) = \mathcal{E}_p(z_{t-k+1:t}), \qquad z_\tau = \mathcal{E}_v(y_\tau), \tag{11}$$

where k is the context-window length. Conditioning on a window rather than a frame is what makes the road context identifiable: as the vehicle advances, geometry that was previously seen far ahead becomes the geometry the vehicle is on, so the window supplies the parallax and motion cues that disambiguate cross-track error, heading error and curvature, which a single frame leaves ambiguous. In practice we realize $\mathcal{E}_p$ convolutionally, with the k frames of the window entered as input channels and a linear curvature head

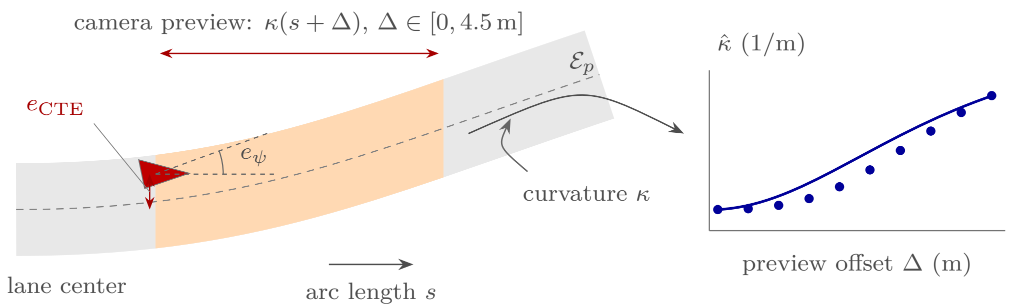

Figure 3: Encoding the road environment. The road-context state $z^r = (e_{\mathrm{CTE}}, e_\psi, \kappa)$ is defined geometrically with respect to the lane center: the cross-track error is the signed lateral offset, the heading error is the angle between the vehicle heading and the road tangent, and κ is the local curvature. For a forward-facing camera the road is visible only over a short preview (shaded), so the physical encoder $\mathcal{E}_p$ aggregates a temporal window of images into a curvature *profile* $\hat{\kappa}(s+\Delta)$ sampled at ten offsets over the first 4.5 m ahead. On the real DonkeyCar this encoding is supervised by privileged track localization at training time and recovered from the front camera alone at test time.

on the pooled features; a self-attention alternative over the same window was evaluated and did not help (Table 2). The encoder produces both components jointly but supervises them separately through (10), ensuring z_t^r carries genuine environmental information rather than being absorbed into the vehicle component.

Privileged supervision for real-world deployment. On the real DonkeyCar, the road context cannot be measured by the deployed car itself. We therefore use *privileged supervision*: during training, an external localization system (a known track map together with the logged car pose, as is also obtainable from an overhead camera) provides ground-truth road context Ξ^r for each frame; at test time this signal is withheld and the deployed car observes only its front camera. The encoder (11) is thus trained to predict, from the onboard image stream alone, the road context that the privileged signal supplied during training. This is precisely a teacher–student / learning-from-privileged-information setup [76]. Because z^r is physically interpretable, the recovered road context can be inspected and compared against the true lane geometry, which we use as a diagnostic in Section 7.

5.3. Predicting the Road Environment

Challenge and motivation. Encoding the road at the current step is necessary but not sufficient. To roll out a trajectory over a long horizon without new observations, the world model must anticipate how the road context itself changes as the vehicle advances: the curve that is ahead now is underfoot a few steps later, and the cross-track and heading errors evolve as the vehicle tracks (or departs from) the lane. A model that predicts only the vehicle state, holding the road context fixed or ignoring it, drifts quickly because it steers the predicted vehicle through a road that no longer matches reality. We therefore make road prediction an explicit, physically interpretable part of the dynamics.

Split dynamics. We use a *split-dynamics* model that advances the vehicle and the road context with separate, physically appropriate transitions:

$$\hat{z^v}_{t+1} = \phi^v_\theta(z^v_t,\ a_t), \tag{12}$$

$$\hat{z^r}_{t+1} = f_r(z^r_t,\ z^v_t,\ \hat{z^v}_{t+1};\ \theta_r). \tag{13}$$

The vehicle transition ϕ^v_θ is a structured kinematic model with a small learned actuation map (below); the road transition f_r is a geometry-driven model (below). The superscript distinguishes it from the conference version's ϕ, which advances the whole state from two previous steps; here only the vehicle branch is advanced, and from one. Note which arguments f_r does and does not take. It consumes the vehicle's *motion over the step*—the state before and after the vehicle update—and not the action: the road context is advected by how far and how the vehicle actually moved, so an action influences the road only through its effect on that motion. This is what makes the second equation a transport rule rather than a second learned dynamics model, and it is the reason $\hat{z^v}_{t+1}$ appears on the right-hand side even though the two transitions are otherwise independent. Splitting the dynamics decouples the fast-varying vehicle motion from the slow-varying road geometry and lets each receive its correct inductive bias.

Structured kinematics for the vehicle. For driving environments we replace the generic second-order dynamics of the conference paper with a wheeled-vehicle kinematic prior. In the body frame this is the classical *bicycle kinematic model*

$$x_{t+1} = x_t + v_t \cos(\psi_t + \beta_t)\,\Delta t, \tag{14}$$

$$y_{t+1} = y_t + v_t \sin(\psi_t + \beta_t)\,\Delta t, \tag{15}$$

$$\psi_{t+1} = \psi_t + \frac{v_t}{l_r}\sin(\beta_t)\,\Delta t, \tag{16}$$

with slip angle $\beta_t = \arctan\left(\frac{l_r}{l_f+l_r}\tan(\delta_{f,t})\right)$, where l_f and l_r are the distances from the center of mass to the front and rear axles (unknown and learned during training), $\delta_{f,t}$ is the steering angle, v_t the longitudinal speed and ψ_t the heading.

Because the road context is itself defined relative to the lane (Section 4.2), we express the vehicle in the same road-aligned frame, which yields the equivalent kinematics (18)–(20) below and lets the two branches of the split dynamics share one geometry. In either frame the *pose* kinematics are analytic; what is unknown is how the commanded action drives the vehicle. We therefore learn only a compact actuation map

$$v_{t+1} = v_t + g_\theta^v(z_t^v, z_t^r, a_t), \qquad \omega_{t+1} = \omega_t + g_\theta^\omega(z_t^v, z_t^r, a_t), \tag{17}$$

where g_θ^v and g_θ^ω are small multilayer perceptrons mapping throttle to longitudinal acceleration and steering to yaw acceleration. Every remaining quantity in the vehicle branch is a physical variable propagated by an analytic equation.

Structured road-context transition. The road-context state $(e_{\text{CTE}}, e_\psi, \kappa)$ evolves slowly and geometrically. As the vehicle moves by the increment implied by its predicted motion, the road ahead is effectively re-indexed: the lane geometry is advected past the vehicle. We therefore model f_r as a geometry-driven transition that (i) updates the cross-track and heading errors from the vehicle's predicted lateral and angular motion relative to the road, and (ii) does *not* integrate the curvature forward at all, but re-reads it from the already-perceived preview at the arc length the vehicle has covered.

Concretely, we work in a road-aligned (Frenet) frame in which the vehicle state is $(s, e_{\text{CTE}}, e_\psi, v, \omega)$, with s the arc length along the lane center, v the longitudinal speed and ω the yaw rate. Writing $\kappa_t = \hat{\kappa}(s_t)$ for the perceived curvature at the current arc length and $\eta_t = 1 - e_{\text{CTE},t}\kappa_t$ for the Jacobian of the road-aligned parameterization, the transition is

$$s_{t+1} = s_t + \frac{v_t \cos e_{\psi,t}}{\eta_t}\,\Delta t, \tag{18}$$

$$e_{\text{CTE},t+1} = e_{\text{CTE},t} + v_t \sin e_{\psi,t}\,\Delta t + \varepsilon_{\theta_r}^d, \tag{19}$$

$$e_{\psi,t+1} = e_{\psi,t} + \left(\omega_t - \kappa_t\,\frac{v_t \cos e_{\psi,t}}{\eta_t}\right)\Delta t + \varepsilon_{\theta_r}^\psi, \tag{20}$$

$$\kappa_{t+1} = \hat{\kappa}(s_{t+1}), \tag{21}$$

where Δt is the sampling interval and $(\varepsilon^d_{\theta_r}, \varepsilon^\psi_{\theta_r})$ is a small learned residual, tightly bounded, that corrects effects the rigid geometric update misses (e.g., imperfect lane tracking). Equations (18)–(20) are analytic; only the residual and the actuation map (17) are learned.

Equation (21) is the crucial one. The curvature is a *lookup into the curvature profile already perceived at the start of the rollout* (Section 5.2), indexed by how far the predicted vehicle has traveled. It is therefore not a free latent that the dynamics must propagate, and it cannot accumulate error over the horizon: the road is *observed* once and then re-indexed, not extrapolated. This is the structural reason the rollout stays bounded, and it is what distinguishes the model from world models that carry the road as an unconstrained latent that drifts and, in turn, corrupts the vehicle branch that consumes it. The preview is long enough for this to be well defined for most of the horizon: over a 100-step rollout the vehicle covers on average 3.1 m of arc length, staying inside the 4.5 m perceived preview in 82% of evaluation windows; past the preview the model holds the last perceived value rather than extrapolating. Because f_r operates on the physically interpretable z^r, the forecasted road context at any horizon is itself interpretable: the model outputs a predicted lane geometry that can be visualized and checked, not an opaque feature.

5.4. Training Procedure

Training follows a staged pipeline. Stage 1 trains the vision autoencoder $(\mathcal{E}_v, \mathcal{D}_v)$ on image reconstruction. Stage 2 trains the physical encoder $\mathcal{E}_p$ to produce the factorized interpretable state, optimizing (10) jointly with a latent reconstruction term. Stage 3 trains the split-dynamics models (ϕ^v_θ for the vehicle and f_r for the road context), with an optional end-to-end fine-tuning pass that updates the encoder and dynamics together to align the encoded state with what the dynamics can predict. Algorithm 1 summarizes the procedure.

6. Experimental Evaluation

6.1. Environments and Evaluation Tiers

We evaluate across three tiers of increasing difficulty. Tier 1 establishes continuity with the conference paper; Tiers 2 and 3 are the two new partially observable driving case studies (Contribution 1).

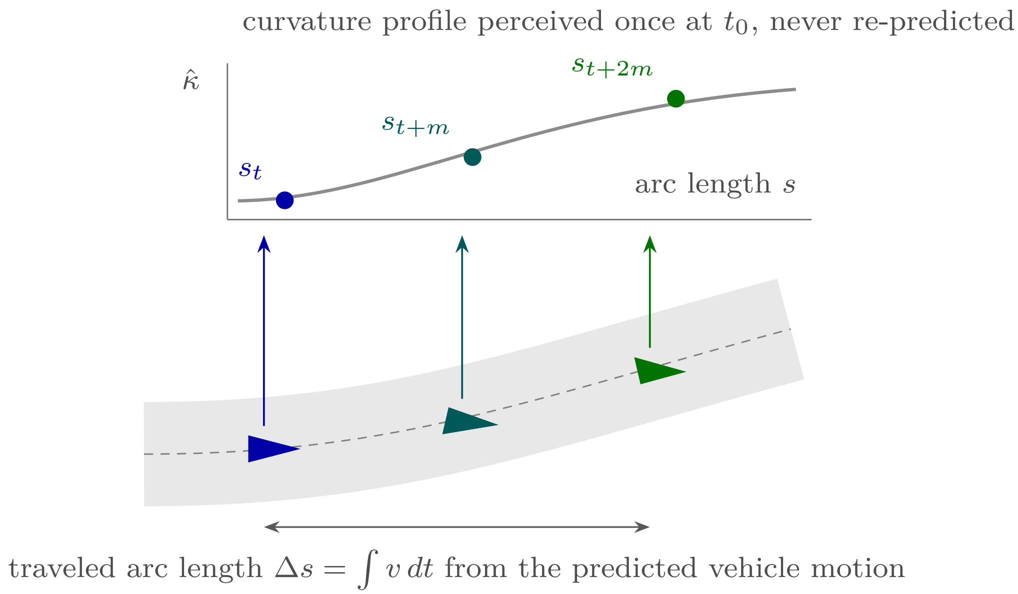

Figure 4: Predicting the road environment. The curvature profile ahead is perceived once, at the start of the rollout; as the vehicle advances, the road context is *advected* past it, i.e. the profile is re-indexed by the arc length the predicted vehicle motion has covered, and the cross-track and heading errors are updated from the vehicle's lateral and angular motion relative to the road. A small learned residual corrects effects the rigid geometric update misses. Because the road is re-indexed rather than propagated as a free latent, it cannot drift over a long horizon, and because z^r stays physically interpretable, the forecasted lane can be inspected at every step.

Tier 1: Fully observable benchmarks. **CartPole** and **Lunar Lander** [81] are retained from the conference paper as controlled settings where the system state is a complete Markov description. These results establish a clean baseline for physically interpretable prediction under simple dynamics.

Tier 2: Partially observable simulated driving (CarRacing). **CarRacing** [81] is a top-down simulated racing environment in which the vehicle navigates a procedurally generated track. The observation is a bird's-eye-view RGB image (native 96×96, resized to 64×64 for the encoder). The vehicle's kinematic state is available from the simulator, but the road geometry (the shape of the upcoming track and the vehicle's relationship to the lane) is not part of the state and must be inferred from the image. Because the camera is top-down, the road ahead is visible but the model must still *encode* it into the physical road context and *predict* its evolution during rollout. This is

Algorithm 1 PIWM training for partially observable driving.

Input: Training set $\mathcal{S}$, loss weights $\lambda_v, \lambda_r, \lambda_{\text{latent}}$, context length k

Train vision autoencoder $(\mathcal{E}_v, \mathcal{D}_v)$ on image reconstruction

for each batch $(y_{1:M}, a_{1:M}, \Xi^v_{1:M}, \Xi^r_{1:M})$ **do**

  $z_t \leftarrow \mathcal{E}_v(y_t)$ for all t

  $(z^v_t, z^r_t) \leftarrow \mathcal{E}_p(z_{t-k+1:t})$ ▷ encode vehicle and road (Sec. 5.2)

  Update $\mathcal{E}_p$ with $\lambda_v \, \mathcal{L}_{\text{interp}}(z^v_t, \Xi^v_t) + \lambda_r \, \mathcal{L}_{\text{interp}}(z^r_t, \Xi^r_t) + \lambda_{\text{latent}} \, \mathcal{L}_{\text{recon}}$

end for

for each batch **do**

  Obtain (z^v_t, z^r_t) from the encoders

  $\hat{z^v}_{t+1} \leftarrow \phi^v_\theta(z^v_t, a_t)$ ▷ bicycle model

  $\hat{z^r}_{t+1} \leftarrow f_r(z^r_t, z^v_t, \hat{z^v}_{t+1}; \theta_r)$ ▷ road prediction (Sec. 5.3)

  Update θ (bicycle) and θ_r (road) on the multi-step rollout loss $\|\hat{z^v}_{t+1} - \hat{\mu}_{\Xi^v_{t+1}}\|^2 + \|\hat{z^r}_{t+1} - \hat{\mu}_{\Xi^r_{t+1}}\|^2$

end for

(Optional) End-to-end fine-tune $\mathcal{E}_p, \phi^v_\theta, f_r$ jointly

return trained model parameters

the controlled setting in which the factorized-state formulation and the split dynamics were developed; the quantitative evaluation reported in this article is on the physical platform of Tier 3, for the reason given in Section 7.2.

Tier 3: Real-world DonkeyCar deployment. The **DonkeyCar (real)** tier evaluates the factorized formulation on a physical RC-scale autonomous car driven on a laboratory track. The car observes its environment through a front-facing camera (the test-time observation, 64×64), and logs wheel-encoder speed, steering, and pose. This is the most demanding tier: the camera is forward-facing, so the road ahead is only partially visible, and the road context cannot be measured by the deployed car. We collect driving trajectories on the physical platform and derive ground-truth road context from external track localization, used as privileged supervision at training time only (Section 5.2); at test time the model infers and rolls out road context from the onboard front-camera stream alone.

6.2. Experimental Setup

Benchmarks (Tier 1). The data-collection and weak-supervision protocol follows the conference paper. For each environment, we collect 60,000 trajectories of at least 50 steps, generated by a mix of random actions and a pretrained neural controller. Weak-supervision labels are synthesized by adding biased uniform noise of relative width $\delta \in \{0\%, 5\%, 10\%\}$ to the ground-truth state, as described in Appendix A.

CarRacing (Tier 2). Driving data are generated with a pretrained controller on procedurally generated tracks; each episode records the onboard image, action, and simulator state. Ground-truth labels for the vehicle pose and for the road context are available directly from the simulator and its track geometry. We use exact ground-truth labels for training (corresponding to $\delta = 0$) and evaluate whether the factorized-state formulation improves long-horizon prediction relative to baselines that use only vehicle-state labels.

DonkeyCar real (Tier 3). We drive the physical DonkeyCar on a closed laboratory track and record synchronized front-camera frames (64×64), control actions (steering, throttle), and vehicle pose, at approximately 22 fps, so that $\Delta t = 1/22\,\mathrm{s}$ and the 100-step evaluation horizon corresponds to about 4.5 s of driving. Trajectories are segmented into clean contiguous laps; brief sensing artifacts (dropped or interpolated frames, pose glitches) are filtered out before training. Ground-truth road context is computed from the known track centerline and the logged pose (privileged supervision), and is withheld at test time. Laps are split into disjoint training and validation sets; all numbers reported below are computed on the 201 held-out validation windows, in real meters.

Architecture and training. The driving experiments use a vision encoder followed by a physical head that produces the factorized state. The head is trained directly against the road-context labels rather than through a reconstruction objective, because road context is available under privileged supervision at training time and nothing has to be learned by reconstructing the image. This is a departure from the conference version's autoencoding variants, so we implemented those as well and measured them here rather than inheriting their ranking (Table 2); on this task the choice does not move the rollout error. The vision encoder maps each 64×64 observation to a compact latent, and the physical encoder aggregates a context window of $T_{\text{ctx}} = 15$ frames ($\approx 0.7\,\mathrm{s}$) to infer the road context from image sequences (Section 5.2). Our rollouts are trained on 16-step horizons and evaluated at up to 100 steps, so the reported horizon is more than six times the training horizon; the baselines are trained on 32-step rollouts, i.e. twice ours, which if anything favors them at the horizons where we claim an advantage. Dynamics are trained with Adam, a cosine learning-rate schedule and early stopping on validation loss. All baselines are trained under identical conditions with the same data, the same optimizer and the same evaluation protocol, and with the batch sizes at which each baseline was tuned; we verified that increasing the batch size degrades these small, latency-bound baselines and would therefore have flattered our model.

6.3. Baselines

We compare against the same families used in the conference paper. **DVBF** [45], **GOKU-net** [43] and **Vid2Param** [59] are physics-aware latent sequence models, and **SINDYc** [60] is a classical dynamics-discovery baseline operating on the interpretable latent. Every baseline is initialized from the ground-truth vehicle state *and* the ground-truth lane waypoints at the first timestep of the window, so each starts from an explicit description of the road that our model has to recover from pixels. They are then run as latent transition models, consuming only actions over the rollout. One asymmetry among them should be stated plainly, because it determines their ordering. Vid2Param additionally infers a latent parameter vector from the image window, since inferring physical parameters from video is the whole of that method; it is consequently the only baseline with any visual input, and it is also the strongest one we measure. DVBF and GOKU-net are run without their own observation pathways—DVBF without its filtering step and GOKU-net without its inference of ODE parameters from the observation sequence—so their numbers should be read as characterizing a *dynamics-only* version of each method rather than the method as published. That reduction was made to hold the comparison at the level of the transition model, but it is a limitation of our evaluation, not a property of those methods, and the gap between our model and them is correspondingly easier to obtain than the gap against Vid2Param. To separate the contribution of *perceiving* the road from that of *representing* it, we also evaluate a privileged **oracle** variant of our own model in which κ is read from the known track map instead of the camera; the gap between the two is the cost of perception.

6.4. Metrics

The primary driving metric is the *position error at rollout step k*: the Euclidean distance, in meters, between the predicted and the true vehicle position after k autoregressive steps, averaged over the $N = 201$ held-out validation windows,

$$E_{xy}(k) = \frac{1}{N}\sum_{i=1}^{N}\left\|\hat{p}_{i,k} - p_{i,k}\right\|_2, \tag{22}$$

where $p_{i,k}$ is the ground-plane position of the vehicle at rollout step k of window i, in meters, and $\hat{p}_{i,k}$ the model's prediction of it. All models are compared in the same planar frame, whatever internal parameterization each uses. We report E_{xy} at $k \in \{25, 50, 100\}$ and plot the full curve $k \mapsto E_{xy}(k)$; all rollouts are initialized from the ground-truth state at $k = 0$ so that the comparison isolates the dynamics rather than the encoder. Because E_{xy} is a distance and not a squared error, its values are directly readable as centimeters of drift. We additionally report the curvature error of the road-context encoder, RMSE_κ in m^{-1}, to evaluate the encoded road directly. For benchmark tasks (Tier 1) we follow the conference protocol and report state RMSE over all state dimensions at a 30-step horizon, together with the relative parameter-recovery error for the environments whose ground-truth physical parameters are known (CartPole, Lunar Lander).

7. Results

7.1. Benchmark Results (Tier 1)

The fully observable benchmark results are carried over from the conference paper [8] and summarized here for completeness. Across CartPole, Lunar Lander and the *simulated* DonkeyCar of the conference paper—which is a different platform from the physical car of Section 7.3—the extrinsic-discrete PIWM variant consistently achieves the lowest and most stable 30-step prediction error, substantially outperforming data-driven and physics-aware baselines; the quantized latent provides strong regularization that improves long-horizon stability. PIWM also recovers the ground-truth physical parameters of CartPole and Lunar Lander with low relative error across all noise levels δ, confirming that the learned representations are genuinely physically grounded rather than merely correlated with the physics. These benchmark results establish that the driving extension is a proper superset of the original framework: on fully observable tasks, where the vehicle state is already Markov, PIWM behaves exactly as before.

7.2. CarRacing (Tier 2)

CarRacing serves in this article as the controlled simulated setting in which the factorized formulation was developed: the road geometry is present in the image but absent from the vehicle state, so the encoder must recover it from pixels even though the top-down view makes the road ahead fully visible. We do not report a quantitative baseline comparison for this tier here. The comparison in an earlier version of this work was run with a vehicle-only dynamics variant rather than the road-context model described in Section 5.3, and re-running it with the present model is left to future work; Section 8 returns to this point. All quantitative claims in this article therefore rest on the physical platform of Section 7.3, where the forward-facing camera makes partial observability genuine rather than nominal.

7.3. Real DonkeyCar Results (Tier 3)

We deploy the learned model on the physical DonkeyCar, where the road context is supervised only during training and must be inferred from the front camera at test time. This is the setting the article is about: the camera sees a short stretch of road, the vehicle state alone is non-Markov, and nothing about the track is available to the deployed model beyond what it can read from its own images.

Long-horizon prediction against baselines. Table 1 and Figure 5 report position error over a 100-step ($\approx$4.5 s) autoregressive rollout under five-fold cross-validation over the 71 recorded episodes: each fold holds out one fifth of them (301–647 evaluation windows, 2483 in total), every model is trained on that fold's training episodes only, and we average within a fold before reporting the mean and standard deviation over the five fold-level means. Comparisons *within* a fold are paired—every model rolls out from the same initial states over the same windows—and the paired statistics below are computed that way. Two regimes are visible, and the crossover between them is the central result. For the first two thirds of the horizon the physics-aware baselines are *ahead* of our model: Vid2Param in particular is nearly four times as accurate at step 25 (0.036 m against 0.141 m). This is expected, and it is a consequence of the comparison being generous to them: each baseline is handed the ground-truth lane waypoints at t_0, so over short horizons it is interpolating from a description of the road that our model had to recover from pixels. Beyond the crossover the ordering inverts and does not invert back: on the fold-mean curves our model is ahead of DVBF from step 64

Table 1: Position error E_{xy} in meters on the physical DonkeyCar, at three rollout horizons, as mean $\pm$ standard deviation over five-fold cross-validation (lower is better). All models are driven by the same actions and are initialized from the ground-truth state; the baselines additionally receive the ground-truth lane waypoints at t_0. Of the baselines only Vid2Param also sees the images—DVBF and GOKU-net are run as dynamics-only models (Section 6.3), which is why Vid2Param leads them. Every model, ours included, keeps the epoch with the best $E_{xy}(100)$; the last column also reports how much each model moves when the held-out episodes change. The map-free row recovers position without consulting the track map at all, reading the road geometry from the perception's shape output instead of indexing a surveyed centerline.

Model	$E_{xy}(25)$	$E_{xy}(50)$	$E_{xy}(100)$
Vid2Param [59]	$\mathbf{0.036 \pm 0.008}$	$\mathbf{0.107 \pm 0.008}$	0.703 ± 0.142
GOKU-net [43]	0.046 ± 0.012	0.131 ± 0.013	0.725 ± 0.107
DVBF [45]	0.061 ± 0.011	0.173 ± 0.031	0.837 ± 0.131
SINDYc [60]	diverges in all 2483 windows of all five folds		
PIWM-Frenet (ours, perceived κ)	0.141 ± 0.009	0.245 ± 0.021	$\mathbf{0.463 \pm 0.045}$
PIWM-Frenet (map-free)	0.146 ± 0.012	0.258 ± 0.023	0.516 ± 0.056

onward, of GOKU-net from step 75 and of Vid2Param from step 82. By step 100 it reaches 0.463 m against 0.703 m for the strongest baseline. Within each fold, where the comparison is paired, our model is ahead at step 100 in *all five* folds against every baseline: by 0.241 m on average against Vid2Param, 0.262 m against GOKU-net and 0.374 m against DVBF. SINDYc diverges in every one of the 2483 evaluation windows of all five folds, so we report it as divergent rather than quoting a value; past the point of divergence the number records only where the rollout was clamped.

The more informative quantity is the spread. Changing which episodes are held out moves our error by ± 0.046 m, against ± 0.192 m for Vid2Param, ± 0.122 m for GOKU-net and ± 0.160 m for DVBF—factors of 2.7 to 4.2. The baselines' long-horizon error depends substantially on which part of the track they were trained on; ours much less so. We regard this as the more durable of the two claims, because it does not rest on any single trained model.

What the crossover means. The two regimes separate two different things a world model can be good at. Before the crossover the task is essentially smooth extrapolation of the recent motion, and a flexible latent sequence model does it well. After the crossover the rollout has left the stretch of road that the initial observation described, and the question becomes whether the

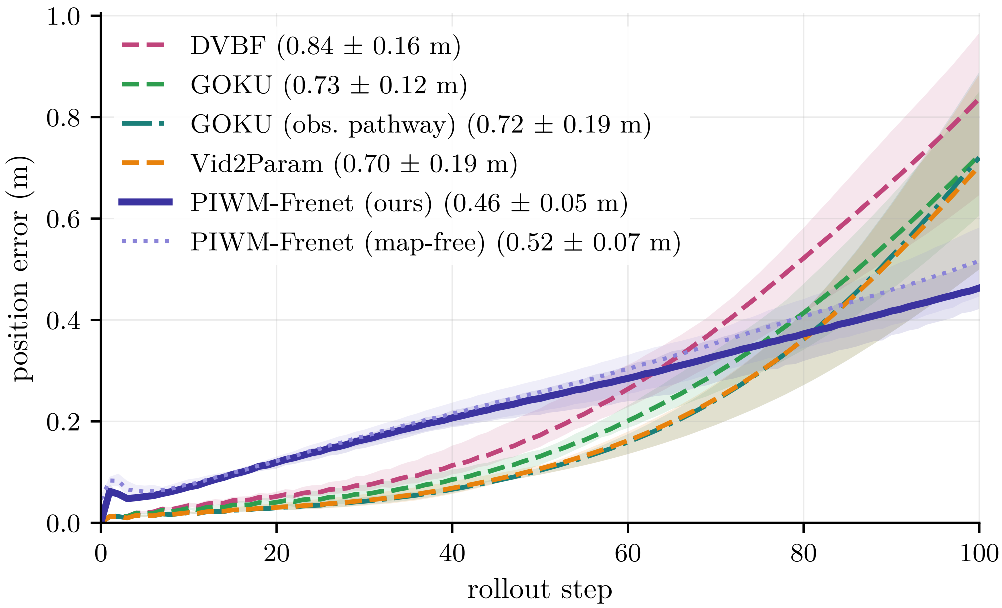

Figure 5: Position error against rollout step on the physical DonkeyCar. Lines are the mean over five folds and the bands span the folds' minimum to maximum, so the width of a band is how much a model depends on which episodes were held out. All rollouts are initialized from the ground-truth state. The baselines lead over short horizons and then enter superlinear growth; the proposed model stays close to linear because the road it rolls out over was observed rather than extrapolated. SINDYc diverges in every window and is omitted.

model knows what road it is now on. A model that carries the road as a free latent must propagate it, and any error in that propagation feeds back into the vehicle branch that consumes it; the result is the compounding growth visible in Figure 5. Our model does not propagate the road at all: by (21) it re-reads the curvature that was already perceived at t_0, indexed by the arc length covered so far (Figure 4). The vehicle travels 3.1 m on average over the full rollout, within the 4.5 m preview in 82% of the evaluation windows; in the remainder the model holds the last perceived value rather than extrapolating. For most of the horizon the road is therefore *observed*, and error in it cannot accumulate.

The cost of perceiving the road. Because the road-context encoder (Figure 3) is the only part of the model that has to look at pixels, we can price it directly by replacing its output with progressively better sources of κ (Table 2).

Feeding the ground-truth curvature profile—perfect perception of exactly the quantity the encoder predicts—yields 0.466 m; the encoder actually used yields 0.463 m, a paired difference of 0.003 m. Reading κ from the known track map instead—privileged information no image-and-action model could have—is *worse*, at 0.505 m; we return to why below. The advantage over the baselines is thus attributable to the *structure* of the representation, not to privileged information or to perception quality.

The table also settles the encoder-architecture question: it does not matter. The ten camera rows span the conference version's design space—directly supervised regressors from scratch and from the pretrained backbone, the two-stage extrinsic autoencoder, the single-encoder intrinsic variant, the discrete VQ–Transformer configuration the conference paper found strongest on fully observable benchmarks—together with plain LSTM and Transformer sequence models as non-physical benchmarks. Across these, curvature error spans a factor of 7.3 (0.122–0.898 m^{-1}) while the 100-step position error moves by 0.017 m, about 4% and less than any single row's spread across folds; paired within folds, no camera encoder differs from the shipped one by more than 0.009 m. The ordering does not even carry over: the most accurate curvature estimate (the Transformer) produces the worst rollout of the ten, and the least accurate (the LSTM, 7 times worse) the second best. This is the clearest evidence we have for the article's central claim. If the rollout depended on continuously estimating the road, an estimate seven times worse would compound over 100 steps. It does not, because the road enters the dynamics as a fixed preview that is re-indexed geometrically (21) rather than integrated: errors in $\hat{\kappa}$ perturb the trajectory but do not accumulate in it.

What does matter is that re-indexing, and the two privileged rows isolate it. Both hold perfect curvature information and differ only in how it is read: the map row looks κ up at the position the rollout has predicted, while the ground-truth profile row reads a preview observed once and indexed by the arc length covered. The second is better by 0.039 m, consistently in all five folds—a larger effect than the entire span of the encoder design space above it. Reading a fixed preview beats consulting a perfect map, because a drift in predicted position sends the map lookup to the wrong part of the track while it merely shifts the preview. The corollary is practical: effort spent sharpening road perception has sharply diminishing returns here, and the remaining error is dominated by the vehicle branch.

Table 2: Where the curvature comes from. All rows use the identical dynamics and differ only in how κ is obtained; five-fold cross-validation, mean $\pm$ standard deviation. RMSE_κ is the encoder's own curvature error in m^{-1} and $E_{xy}(100)$ the resulting 100-step position error in meters. The first two rows are privileged and not deployable; the remaining ten read the front camera alone and span the conference version's encoder design space together with its two non-physical sequence benchmarks. Curvature accuracy varies by a factor of 7.3 across the camera rows while the rollout error varies by 4%—less than any single row moves between folds—and the ordering does not carry over: the best curvature estimate gives the worst rollout.

Source of κ	Input	RMSE_κ	$E_{xy}(100)$
Known track map	map + pose	—	0.505 ± 0.044
Ground-truth profile	true κ	—	0.466 ± 0.049
Extrinsic autoencoder	camera	0.193	$\mathbf{0.454 \pm 0.061}$
LSTM	camera	0.898	0.456 ± 0.112
Intrinsic autoencoder	camera	0.136	0.460 ± 0.047
CNN, frozen backbone	camera	0.247	0.461 ± 0.063
CNN, warm-started backbone	camera	0.148	0.462 ± 0.056
CNN, from scratch	**camera**	**0.154**	$\mathbf{0.463 \pm 0.046}$
VQ + Transformer	camera	0.219	0.463 ± 0.056
Transformer	camera	**0.122**	0.471 ± 0.052

Robustness to weak supervision. All results above are trained with exact road-context labels. The conference formulation, however, supervises the interpretable state only through noisy distributions, and the physical platform is exactly where such labels are hardest to obtain: they come from an external localization system whose accuracy is finite. We therefore repeat the training under the conference noise model of Appendix A, injecting biased uniform label noise of relative width $\delta \in \{0\%, 5\%, 10\%\}$ (Table 3, Figure 6). The two model families respond in opposite directions. Every baseline degrades as the labels get noisier. Our model improves, monotonically, from 0.463 m at $\delta = 0$ to 0.319 m at $\delta = 10\%$—a 31% reduction, while Vid2Param worsens by 33% over the same range. The trend holds in all five folds, and our model's intervals at $\delta = 0$ and $\delta = 10\%$ do not overlap.

This is not a property of the conference formulation's particular noise. Replacing it with a zero-mean Gaussian of the same per-dimension magnitude, holding everything else fixed, gives 0.313 m against 0.319 m—a paired difference of -0.005 m whose sign is not consistent across folds. Label noise at this magnitude regularizes the structured model, and any zero-mean noise of that magnitude does it. The finding is the *asymmetry* between the model

Table 3: Weak-supervision noise on the physical DonkeyCar. Position error $E_{xy}(100)$ in meters when the labels used for training are corrupted by biased uniform noise of relative width δ (Appendix A), as mean ± standard deviation over five folds. The structured model improves as the labels degrade; every baseline gets worse. The last row replaces that noise with a zero-mean Gaussian of matched per-dimension magnitude, which reproduces the gain and shows it to be ordinary regularization rather than a property of the noise model.

Model	$\delta = 0\%$	$\delta = 5\%$	$\delta = 10\%$
PIWM-Frenet (ours)	0.463 ± 0.045	0.452 ± 0.035	$\mathbf{0.319 \pm 0.044}$
PIWM-Frenet (map-free)	0.516 ± 0.056	0.506 ± 0.049	0.390 ± 0.036
Vid2Param	0.703 ± 0.142	0.918 ± 0.189	0.938 ± 0.211
GOKU-net	0.725 ± 0.107	0.896 ± 0.152	0.864 ± 0.142
DVBF	0.837 ± 0.131	1.067 ± 0.188	1.075 ± 0.177
matched-magnitude Gaussian control, ours at $\delta = 10\%$: 0.313 ± 0.041			

families, not the form of the noise. Consistent with this, the baselines' selected epochs move sharply earlier as δ grows—several select the fifth epoch at $\delta = 10\%$—which is what fitting the label noise almost immediately looks like.

Robustness to a corrupted initial state. The study above perturbs the *supervision*; a complementary question is what happens when the state the rollout starts from is itself wrong, as it would be under an imperfect state estimator at deployment. We therefore repeat the evaluation with Gaussian noise of standard deviation σ (in units of each state variable's own standard deviation) added to the initial state, 10 draws per window (Figure 7 (a)). This is a *test-time* perturbation of the state and should not be confused with the δ *training-label* noise of Table 3; the two probe different failure modes. The ranking is unchanged at every level: at $\sigma = 1.5$ our model is at 0.749 m against 0.889 m (Vid2Param), 1.186 m (GOKU-net) and 1.393 m (DVBF). We note that our relative degradation over this range (2.8×) is in fact slightly larger than the baselines' (2.3–2.5×), which is what one expects when a method starts from the smallest error: the absolute gap narrows with noise, but it does not close.

Because every state variable is a physical quantity, we can also ask *which* one the rollout is sensitive to, rather than only how sensitive it is overall (Figure 7 (b)). Perturbing the speed v is by far the most damaging—0.4 m/s of noise roughly doubles the 100-step error—while the yaw rate ω is almost

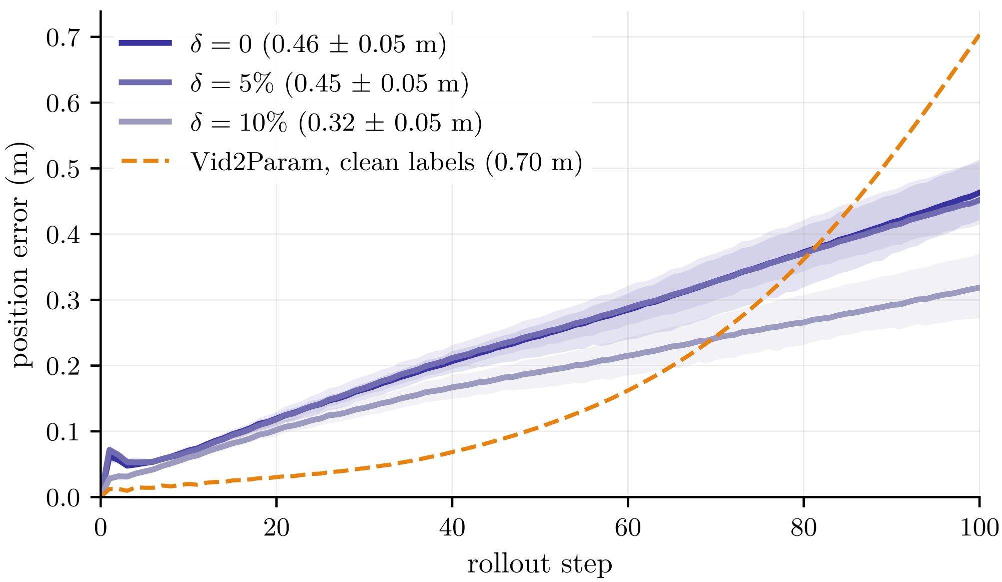

Figure 6: Robustness to corrupted supervision. Position error against rollout step for our model when the training labels carry biased uniform noise of relative width δ; lines are fold means and bands span the five folds. The dashed reference is the strongest baseline trained on *clean* labels. Noisier labels move our curves down, not up.

inconsequential over an order of magnitude larger range, and the road-context variables e_{CTE} and e_ψ sit in between. This is a diagnostic an opaque latent state cannot provide, and it is actionable: it says that longitudinal state estimation, not heading estimation, is where sensing effort should go on this platform.

8. Discussion

Bounded rollout comes from not having to predict the road. The horizon at which our model overtakes the baselines (Figure 5) is the horizon at which the initial observation stops describing the road the vehicle is on. The structural answer this article proposes is to make the road not a predicted quantity at all but an *observed* one that is re-indexed by the vehicle's own predicted motion (21). Vehicle state alone is non-Markov, so the road context must be *encoded*—but once encoded over a preview that outlasts the horizon, it need not be extrapolated, and that is what keeps the error bounded.

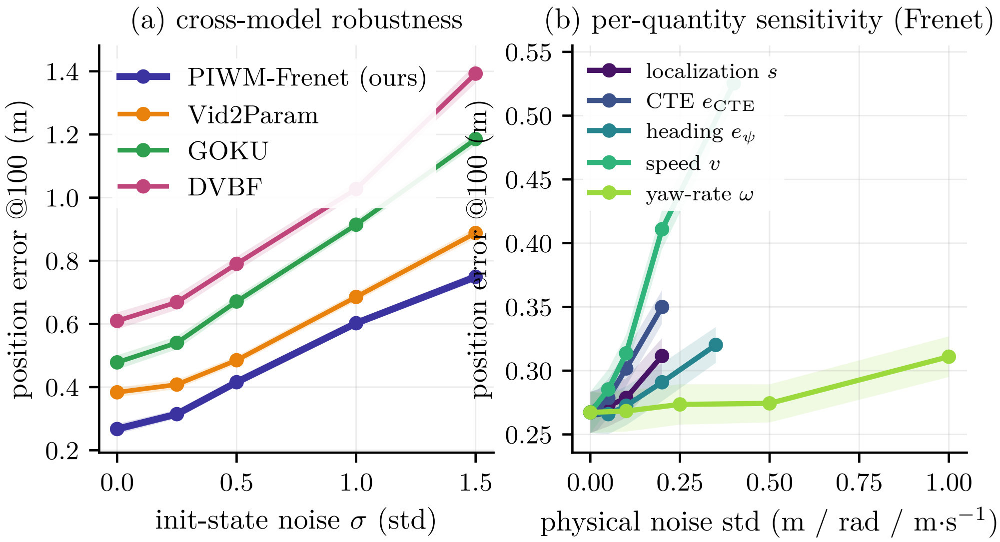

Figure 7: Robustness to a corrupted *initial state* (test-time perturbation; distinct from the training-label noise of Table 3). (a) 100-step position error against the noise scale σ applied to all state variables; the ranking is preserved throughout. (b) Sensitivity of our model to noise on each physical state variable separately. Speed dominates; yaw rate is nearly irrelevant. Shaded bands are standard errors over the 201 validation windows.

Structure, not information. It would be easy to attribute our long-horizon advantage to the privileged supervision used at training time. Table 2 rules this out. Replacing the camera-derived curvature by a lookup into the true track map—the strongest form of privileged information available—makes the 100-step error *worse*, from 0.463 m to 0.505 m, consistently in all five folds, and feeding the ground-truth curvature profile moves it by 0.003 m. The 0.24 m separating our model from the strongest baseline is therefore not privileged information on our side; the baselines were in fact handed ground-truth lane waypoints at t_0, which our model never sees. The baseline ordering itself points the same way: the single baseline that retains a visual pathway, Vid2Param, is also the best of them, which is the same lesson the road-context encoder teaches. What separates them is that our state variables are road-aligned physical quantities propagated by analytic geometry, rather than latent coordinates propagated by a learned map.

Interpretability of the forecasted environment. Because the road context is physically interpretable, the forecasted road, not just the forecasted vehicle, can be inspected and monitored. The predicted curvature profile can be compared against the true track geometry at every step—the encoder's own error, $\mathrm{RMSE}_\kappa = 0.154\,\mathrm{m}^{-1}$, is measured in the physical unit of curvature—providing a diagnostic that opaque world models do not offer. This is directly useful for safety monitoring: a forecasted road that departs from plausible geometry is a detectable failure signal, available before the predicted trajectory itself looks wrong. The same property makes the model's failure modes addressable rather than merely observable: because the state is physical, the sensitivity study of Figure 7 (b) attributes the rollout's fragility to a specific quantity—longitudinal speed—and is silent about another—yaw rate—which turns a robustness result into a concrete instruction about where sensing effort belongs. A latent-state model can report that it is sensitive to initialization; it cannot say to *what*.

Privileged training for real deployment. The real DonkeyCar uses privileged road-context supervision at training and withholds it at test. This reflects practical deployability: external localization (a mapped track, or an overhead view) is available in instrumented training environments but not on a freely deployed car. The fact that the learned encoding transfers to the unaided setting is what makes the approach viable on real hardware.

Limitations. First, the road-context variables are specified manually for each environment; a more general approach might discover the decomposition automatically. Second, the vehicle kinematics are a slip-free approximation that ignores tire dynamics, which may limit accuracy at high speeds or on low-friction surfaces; the physical platform used here operates well inside the linear regime, so this assumption is untested at the limit. Third, the argument that the road need not be extrapolated holds only while the horizon stays inside the perceived preview. On our platform the vehicle covers 3.1 m on average per 100-step rollout, but the 4.5 m preview contains the full rollout in only 82% of the evaluation windows; in the rest the model holds the last perceived curvature. A faster vehicle or a longer horizon would leave the preview more often, and the model would then have to either re-perceive or genuinely extrapolate the road; we have not characterized that regime. Fourth, privileged supervision requires an instrumented training environment to obtain ground-truth road context; reducing this requirement (e.g., via self-supervised road estimation) is open. Fifth, every model's epoch is selected on the same held-out windows the tables report. We apply that rule symmetrically, which removes an asymmetry present in an earlier version of this evaluation, but symmetric selection is not unbiased selection and the absolute values should be read with that in mind. It also differs from the conference version, which stopped every model early on validation loss. Sixth, the quantitative evaluation reported here is confined to the physical platform: the CarRacing comparison referred to in Section 7.2 was run with a vehicle-only dynamics variant and has not been repeated with the road-context model, so this article does not claim a simulated-domain result. Seventh, two of the three baselines are evaluated in a reduced form. Holding the comparison at the level of the transition model meant running DVBF without its filtering and GOKU-net without its inference of ODE parameters from observations, leaving Vid2Param as the only baseline with a visual pathway. This understates DVBF and GOKU-net, and it is the most likely explanation for their ordering relative to Vid2Param reversing between our simulated and physical experiments; a comparison against full implementations of both would be a stronger test than the one we report. Finally, the method currently assumes a single vehicle and a single lane; multi-agent and multi-lane settings are future work.

Future directions. Natural next steps include scaling PIWM to open-world driving with richer road topology, multiple agents, and additional sensor modalities (LiDAR, radar); treating other vehicles as additional interpretable road-context variables to handle interactions and occlusions; and discovering the road-context structure from data via symbolic regression or causal structure learning, which would reduce the need for environment-specific design.

9. Conclusion

This article extended the Physically Interpretable World Model (PIWM) from fully observable benchmarks to partially observable driving. The central observation is that driving is partially observable: the vehicle's physical state alone is non-Markov with respect to future trajectory, because the road context (lane geometry and the vehicle's relationship to it) is absent from any standard state representation and only partially visible to an onboard camera. We addressed this along three axes. We introduced two partially observable driving case studies, a simulated CarRacing track and a real physical DonkeyCar. We introduced a physically interpretable *encoding* of the road environment, factorizing the interpretable state into a vehicle-relative component and a road-context component inferred from images, including under privileged supervision for real-world deployment. And we introduced a physically interpretable *prediction* of the road environment: a split-dynamics model in which the vehicle advances under structured kinematics while the road is advected past it, re-indexed by the distance the vehicle has covered rather than propagated as a free latent. Experiments on the real DonkeyCar confirm that this treatment of the external environment, not merely of the vehicle, is what keeps long-horizon prediction bounded under partial observability: at 100 steps and under five-fold cross-validation the model reaches 0.463 m against 0.703 m for the strongest baseline, ahead in every fold and with a fraction of every baseline's sensitivity to which episodes are held out; under label noise of up to 10% it *improves* while every baseline degrades, and the original benchmark results are retained unchanged. Replacing perceived curvature by a privileged map lookup makes the result worse, and perfect perception moves it by 3 mm, so the advantage is attributable to the structure of the representation rather than to the information available to it. Physically interpretable world modeling for driving thus requires aligning latent representations with physical variables *and* decomposing those representations according to the semantic structure of the task, so that the external environment is both encoded and predicted in physically meaningful terms.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

CRediT Authorship Contribution Statement

[CRediT ROLES — TO BE COMPLETED BEFORE SUBMISSION.]

Data Availability

The DonkeyCar trajectories, the derived road-context labels and the code used to train and evaluate the models will be made available in a public repository upon publication.

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Appendix A. Weak Supervision Noise Model

The weak supervision in Tier 1 benchmark experiments follows the noise model from the conference paper. For the i-th state component with valid range size $\mathcal{X}_i$, we generate a randomly shifted center $\tilde{\mu}_i = x_i + \Delta_i$, $\Delta_i \sim \mathrm{Unif}[-\frac{1}{2}\delta|\mathcal{X}_i|, \frac{1}{2}\delta|\mathcal{X}_i|]$, and draw the supervisory distribution as $p_i(x) = \mathrm{Unif}[\tilde{\mu}_i - \frac{1}{2}\delta|\mathcal{X}_i|,\ \tilde{\mu}_i + \frac{1}{2}\delta|\mathcal{X}_i|]$. At each timestep, $L = 50$ samples are drawn to form the proxy supervision set Ξ_t. The same noise model is applied to the road-context supervision Ξ^r on the physical DonkeyCar in the robustness study of Table 3, where δ scales the valid range of each road-context component. Unless stated otherwise, the Tier 3 results reported in Section 7.3 use exact ground-truth labels at training time ($\delta = 0$), reflecting the availability of training-time track localization; at test time the road context is withheld in every case and inferred from images.

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